

LEMMA: Laplacian pyramids for Efficient Marine SeMantic Segmentation

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Abstract

Semantic segmentation in marine environments is crucial for the autonomous navigation of unmanned surface vessels (USVs) and coastal Earth Observation events such as oil spills. However, existing methods, often relying on deep CNNs and transformer-based architectures, face challenges in deployment due to their high computational costs and resource-intensive nature. These limitations hinder the practicality of real-time, low-cost applications in real-world marine settings.

To address this, we propose LEMMA, a lightweight semantic segmentation model designed specifically for accurate remote sensing segmentation under resource constraints. The proposed architecture leverages Laplacian Pyramids to enhance edge recognition, a critical component for effective feature extraction in complex marine environments for disaster response, environmental surveillance, and coastal monitoring. By integrating edge information early in the feature extraction process, LEMMA eliminates the need for computationally expensive feature map computations in deeper network layers, drastically reducing model size, complexity, and inference time. LEMMA achieves an mIOU of 98.97% on the Mastr-1325 dataset and 93.42% on the Oil Spill dataset, effectively demonstrating performance across datasets captured from diverse platforms while reducing trainable parameters and computational requirements by up to 71x, GFLOPs by up to 88.5%, and inference time by up to 84.65%, as compared to existing models. Experimental results highlight its effectiveness and real-world applicability in the Marine Computer Vision.

1. Introduction

Earth observation (EO) from Unmanned Aerial Vehicles (UAV) has become a cornerstone of large-scale environmental monitoring, enabling persistent surveillance of coastal ecosystems, maritime infrastructure, and pollution events such as oil spills and chemical discharges [47] [35].

High-resolution UAV-based EO imagery is increasingly used for rapid disaster response, coastal zone management, and environmental impact assessment due to its low deployment cost, flexible coverage, and high spatial resolution compared to satellite-based sensing [35] [46]. However, EO segmentation in maritime and coastal regions remains particularly challenging because of strong specular reflections, low inter-class contrast between water and thin surface films, atmospheric illumination variations, and dynamic surface textures caused by waves and wind [47] [57]. These challenges impose strict requirements on segmentation models to be not only accurate but also computationally efficient for deployment on resource-constrained aerial platforms used in real-world EO missions. Consequently, there is a growing demand for lightweight, edge-aware semantic segmentation frameworks that can deliver reliable geospatial delineation in complex EO scenarios while maintaining real-time performance.

Semantic segmentation enables pixel-level understanding for various real-world scenarios, and finds applications in UAV domains, autonomous driving systems [48], robotic vision [33], unmanned surface vehicles (USVs), and oil spill identification from aerial drone imagery. These tasks are relevant to operational hazards and environmental damage [31], [19]. Traditional state-of-the-art (SOTA) segmentation models are typically resource-intensive [3], requiring significant memory and computational power, making them impractical for deployment on edge devices such as USVs or drones. The growing demand for real-time, on-device processing has thus driven the need for lightweight, efficient models capable of delivering high performance in resource-constrained environments [52].

The domain of marine semantic segmentation has evolved through diverse architectural paradigms, encompassing both general-purpose and domain-specific approaches. Early works adapted terrestrial segmentation networks like FCN [31], and SegNet [2] to maritime environments with limited success. More sophisticated architectures emerged with DeepLabv3 [9] and PSPNet [56], leveraging atrous convolutions and pyramid pooling for improved contextual understanding. Maritime-specific archi-

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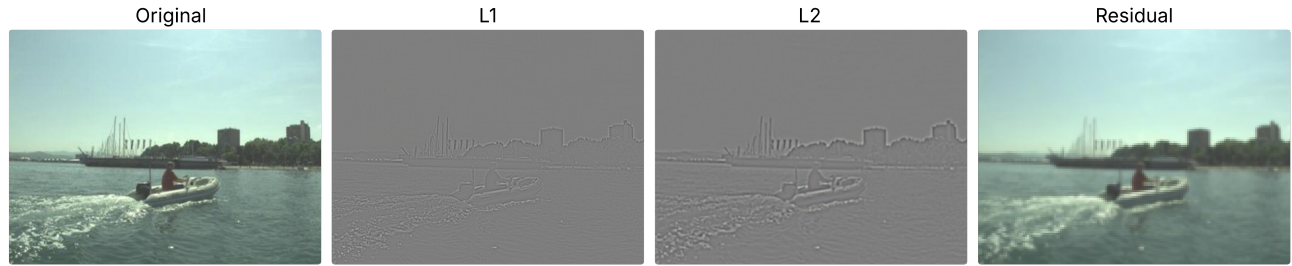


Figure 1. Visualization of the Laplacian pyramid of depth 2 of an RGB image. Vital spatial and structural information is present at each level, notably in the edges available at different resolutions. Here, L_1 , L_2 and Residual (L_3) refer to the first, second and last layer of the decomposed Laplacian Pyramid.

textures such as WaSR [3] and WaSR-T incorporated temporal consistency and transformer modules, while MUNet [1] introduced multi-level attention mechanisms for maritime obstacle detection. The proposed methodology diverges from these approaches by employing a Laplacian pyramid decomposition chain with residual blocks at each level for efficient feature extraction, drastically reducing the computational footprint with up to 71x fewer parameters while maintaining competitive performance across diverse maritime datasets from both surface and aerial perspectives.

Our motivation in employing Laplacian pyramids lies in the edge information returned in a one-shot manner by the decomposition of the pyramid. As visible in Fig. 1, the edges of the obstacles (in this example) are visible at various resolutions and are easy to identify in the layers of the pyramid. This edge information allows for multi-level feature extraction, leading to superior results across images from diverse capturing platforms, like drone and USV data, at a much lower computational cost. This reduction in computational complexity enables the model to run efficiently on edge devices without compromising accuracy, making it ideal for real-world applications. The lightweight design of the proposed approach (LEMMA) bridges the gap between computational constraints and the need for precise segmentation in critical scenarios [25]. Our key contributions are as follows.

- Adapt the Laplacian Pyramid to extract vital edge information, enabling efficient and precise marine semantic segmentation.
- Validate the proposed methodology on two challenging and diverse tasks - obstacle segmentation in USV data and Oil Spill segmentation captured from aerial drones - demonstrating its broad applicability in real-world scenarios and deployment.
- Demonstrate SOTA results while reducing model parameters by up to 71x, GFLOPs by up to 88.5%, and inference time by up to 84.65%, showcasing suitability for deployment on resource-constrained platforms like drones,

UAVs, and USVs.

Unlike prior applications of Laplacian pyramids, which focus on medical or terrestrial contexts, our work adapts the decomposition mechanism for low-cost, edge-aware segmentation in marine environments. We introduce a three-branch residual framework specifically designed to work with pyramid-level edge cues, enhancing thin-boundary prediction without heavy post-processing.

2. Related Work

In this study, previous works relevant to Laplacian pyramids and Marine segmentation have been summarised. Laplacian pyramids are a traditional computer vision technique which finds its applications in various tasks, like segmentation [20], image generation [14], super-resolution [22], style transfer [27], etc.

2.1. Laplacian Pyramid-Based Feature Extraction for Semantic Segmentation

Laplacian pyramids have been widely employed in computer vision for multi-scale feature extraction due to their capability of preserving fine-grained edge details while enabling hierarchical image analysis. These methods are particularly effective in enhancing segmentation performance by capturing both low-frequency structural information and high-frequency edge details, which are essential for distinguishing fine boundaries in complex scenes.

The concept of Laplacian pyramid decomposition was first introduced as a multi-resolution analysis method that decomposes an image into progressively lower-resolution representations while maintaining high-frequency details [5]. This hierarchical approach allows for efficient feature extraction and reconstruction, making it highly effective for segmentation tasks where both local and global contextual information are critical. Paris et al. [38] extended this concept by introducing Local Laplacian Filters, which enhance contrast and detail preservation using Laplacian pyramids in an edge-aware manner. Their method allows for fine-scale

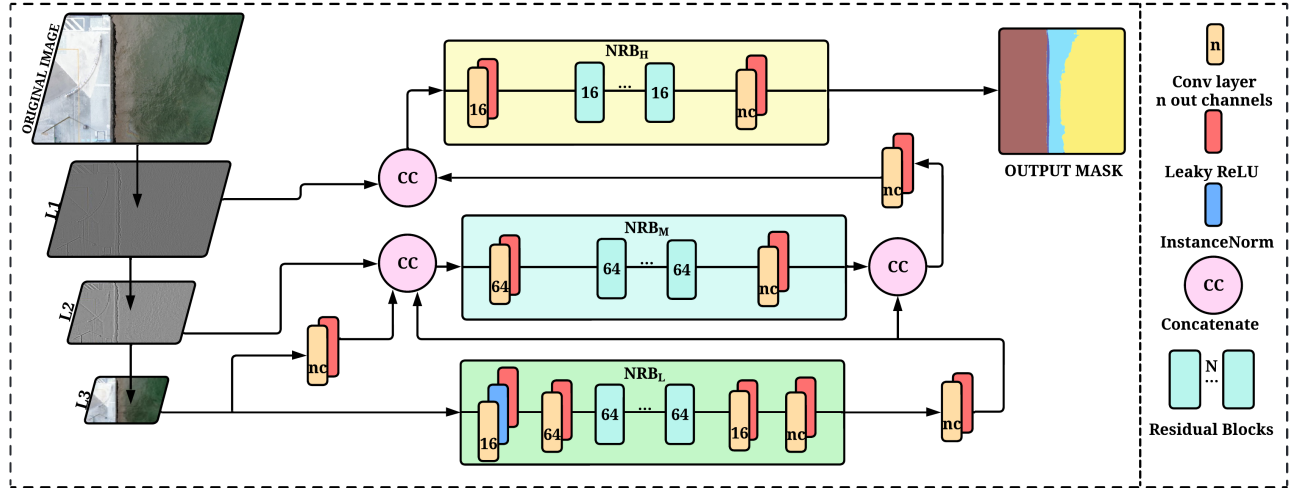


Figure 2. A schematic overview of LEMMA. The sections highlighted in green, blue, and yellow symbolize the LFB, MFB, and HFB, respectively. L_1 , L_2 , and Residual (L_3) represent the three layers of the decomposed Laplacian pyramid. The 'nc' for each dataset is the total number of classes in the dataset. Each residual block chain has the corresponding number of blocks, as explained in Section 3.

control of contrast enhancement without introducing halo artifacts, making it suitable for edge-sensitive applications like semantic segmentation.

2.2. Marine Segmentation

Marine semantic segmentation as a computer vision task involves pixel-level classification of underwater imagery to distinguish marine objects, organisms, and terrain features from their surroundings [49]. Recent advances in deep learning have introduced several state-of-the-art (SOTA) models designed specifically for the challenges of marine perception, such as varying water conditions, occlusions, and real-time processing constraints.

WaSRNet [3] is a maritime segmentation network designed to handle water variability and false positives in USV navigation. It employs an encoder-decoder architecture with feature fusion mechanisms to improve segmentation in cluttered water scenes. By incorporating spatial attention and multi-level feature aggregation, WaSRNet effectively reduces false positives in wave-heavy environments, making it highly effective for maritime navigation in dynamic conditions. However, its encoder-decoder design requires 71.4 million parameters, limiting deployment on edge devices.

Building upon this, BEMSNet [55] enhances boundary segmentation by introducing boundary-aware loss functions and edge-refining modules, achieving SOTA performance on various datasets. While it improves contour accuracy for thin structures like buoys and ship railings, the added boundary enhancement modules increase computational complexity, maintaining a parameter count of 71.4 million. This underscores a critical trade-off between preci-

sion and efficiency in safety-critical applications.

A Lightweight Dual-Branch Semantic Segmentation Network for Ship Navigation [17] proposes separate pathways for low-level and high-level feature extraction, reducing computational overhead compared to single-stream architectures. While this approach achieves real-time processing with 31 million parameters, its dual-branch design still struggles to balance accuracy and speed in highly dynamic marine environments with rapid lighting and surface variations.

The marine segmentation landscape reveals a consistent pattern: Although specialised domain methods achieve high accuracy, none effectively addresses the dual requirements of maintaining precision while enabling real-time deployment on resource-constrained marine platforms. Traditional approaches require tens of millions of parameters and hundreds of GFLOPS, creating a substantial computational barrier for practical marine deployment.

3. Methodology

This section details the architecture of the proposed methodology, LEMMA, a novel edge-recognition-based lightweight model employing Laplacian pyramids to bypass heavy feature map computation for Marine semantic segmentation. Drawing upon established research, we incorporate multi-scale processing with residual connections and feature concatenation across different levels as fundamental components of our design. As depicted in Fig. 2, the original image is decomposed into a Laplacian pyramid of depth 3, resulting in three layers referred to as L_1 , L_2 and L_3 . Features are extracted from these layers at various scales,

which undergo further processing in the High-level Feature Branch, Middle-level Feature Branch and Low-level Feature Branch, respectively.

3.1. Low-level Feature Branch

The input to the Low-level Feature Branch (LFB) is L_3 . Before being processed by the LFB, L_3 undergoes a convolution operation, followed by a leaky ReLU activation to be concatenated with L_2 , before L_2 is processed by the Middle-level Feature Branch.

The LFB consists of various Convolution layers (conv), InstanceNorm, and Leaky ReLU activations. A chain of residual blocks is placed between these modules. These residual blocks consist of a conv layer, leaky ReLU, and another conv layer with a residual connection. The kernel size of 3 and the stride of 1 are constant throughout the architecture. NRB_L denotes the number of these residual blocks. The conv layers before the residual block chain bring the number of channels of the feature map to 64 channels. By integrating the low-level details extracted by the LTB with the higher-level features, the network can construct a more comprehensive feature representation, setting the stage for advanced processing in subsequent network stages. Hence, the feature extraction can be expressed as:

$$\begin{aligned} L_{3f} &= \text{LFB}(L_3) \\ L'_3 &= \text{LeakyReLU}(\text{TransposeConv}(L_3)) \end{aligned} \quad (1)$$

Additionally, before L_3 is processed by the LFB, it is upsampled using a transpose convolution and leaky ReLU activation (L'_3) and concatenated with L_2 and L_{3f} . This is implemented to retain the raw information present at this scale of the pyramid.

3.2. Middle-level Feature Branch

The Middle-level Feature Branch (MFB) receives the concatenation of L_{3f} , L'_3 and L_2 . This feature map is then processed by residual blocks between convolution and leaky ReLU layers. NRB_M indicates the number of these residual blocks.

The MFB is responsible for refining and distilling key features received from L_2 and L_3 . The accentuated edge information and fine details derived from Laplacian pyramids (L_2) help to bypass heavy feature map computation. These derived details allow the model to avoid an excessive number of parameters. The MFB extracts spatial and structural information by processing and refining the joint feature maps of the LFB and the L_2 layer of the pyramid. This advanced feature representation is crucial for the network's subsequent stages, where such high-level conceptualizations of the image are vital for advanced visual tasks.

The output of the conv and Leaky ReLU module placed after the residual blocks is further concatenated with the

output of the LFB. This feature map now contains information refined from the Middle and Lower branches, thereby retaining structural awareness at different scales. This is subsequently passed through another conv and leaky ReLU module and concatenated with L_3 to finally be processed by the High-level Feature Branch. Hence, the processing in the branch can be formulated as:

$$\begin{aligned} L_{\text{concat}} &= [L_2; L_{3f}; L'_3] \\ L_{\text{MFB}} &= \text{MFB}(L_{\text{concat}}) \\ L_{\text{CC}} &= [L_{\text{MFB}}; L_{3f}] \\ L_{\text{out}} &= \text{LeakyReLU}(\text{TransposeConv}(L_{\text{CC}})) \end{aligned} \quad (2)$$

where $[\cdot]$ represents concatenation.

3.3. High-level Feature Branch

The High-level Feature Branch (HFB) carries out further refinement and considers the intricacies present in L_3 , the highest-resolution Laplacian layer. The information extracted from L_3 , along with the merged features of lower branches, allows for accurate reconstruction, taking into account spatial and structural features, enabling swift reconstruction. The HFB starts with a conv layer and leaky ReLU, followed up by NRB_H number of residual blocks. Finally, a conv layer and leaky ReLU activation follow to generate the mask. Following a structure similar to the MFB and LFB but only processing features of 16 channels instead of 64 channels, the HFB ensures lower GFLOPs while reconstructing a highly accurate mask swiftly and inexpensively. The feature processing is formally represented as:

$$\begin{aligned} L_{\text{CC}} &= [L_1; L_{\text{out}}] \\ M_{\text{final}} &= \text{HFB}(L_{\text{CC}}) \end{aligned} \quad (3)$$

where $[\cdot]$ represents concatenation, and M_{final} is the final multiclass segmentation mask.

4. Experimental Setup and Results

To evaluate the proposed methodology, comprehensive evaluations were conducted across two distinct marine segmentation datasets, demonstrating its robustness and generalizability under varied operational paradigms :

Oil Spill Drone Dataset [42]: This dataset focuses on ecological monitoring with 847 high-resolution UAV images of resolution 1920×1080, of port environments containing oil spills, water surfaces, vessels, and shoreline structures. The dataset emphasizes fine boundary detection between spilt hydrocarbons and water surfaces, with per-pixel annotations verified by environmental scientists. A stratified 75-25 split considering the spatial distribution of spills to prevent overlapping regions between sets is implemented. The aerial perspective and thin oil sheens present

Model	mIoU	#Params (M)	GFLOPs	Inference Time (ms)
DeepLabv3 + Xception [10]	80.65	54.71	124.65	29.83
ERFNet [40]	81.68	2.06	22.16	15.07
Pretrained SegNet [2]	81.80	29.45	241.29	22.31
CFPNet [32]	82.28	0.54	5.85	35.61
ERFNetv2 [34]	82.45	1.23	4.18	18.33
DABNet [26]	82.64	0.75	7.76	15.59
LETNet [53]	83.18	0.94	10.63	73.15
UISSNet + ResNet50 [21]	83.56	75.42	256.59	35.15
UISSNet + ResNet18 [21]	83.91	44.11	195.15	24.94
HRNet-w18 [23]	84.11	29.54	67.90	33.85
LGCNet [30]	84.14	0.69	9.05	28.67
Pretrained UNet [41] + VGG16 [43]	86.01	24.89	339.01	48.29
WODIS [11]	91.30	89.50	-	23.15
PSPNet [56]	93.46	66.0	8.91	5.94
Fast SCNN [39]	93.50	1.36	-	14.82
DDRNet-s [37]	94.50	17.05	-	12.66
Segmenter [44]	94.80	-	-	18.73
LDANet [13]	96.16	18.53	-	12.35
LightWeight [46]	96.81	-	-	-
DeepLabv3 [9]	97.67	48.0	123.11	47.55
HRNet [45]	97.87	63.0	-	-
ShorelineNet [54]	98.75	6.5	-	-
SGAF [12]	98.88	63.0	-	-
WaSR-T [58]	99.80	71.4	133.80	-
BEMRF-Net [6]	99.91	71.40	156.0	-
LEMMA (Ours)	98.97	1.07	17.83	7.3

Table 1. Comparison of various models with LEMMA on **mIoU** (in percent) across MaStr1325 [4] and Oil Spill Drone [42] datasets. Parameters, GFLOPs, and inference time are provided for both ResNet-50 and ResNet-101 backbones where applicable. Results for LEMMA are of residual block (Section 3) configurations 7,7,1 and 6,7,4 respectively (separated by forward slashes for Mastr1325 and Oil Spill). Values for SOTA comparisons separated by slashes refer to R50/R101 variants. Unreported values in original works or lack of publically available weights have been left blank.

unique challenges compared to USV-based imagery. This dataset is leveraged as the coastal RGB Earth observation dataset for this study.

Marine Semantic Segmentation Training Dataset (**MaStr1325**) [4]: This benchmark contains 1,325 images of resolution 512×384 captured from unmanned surface vehicles (USVs) operating in coastal environments over two years. The dataset spans five object categories critical for autonomous navigation: water, sky, vessels, obstacles, and animals. Following previous work [3], an 80-20 train-test split is employed with temporal stratification to prevent

leakage between similar scenes captured in sequence. The maritime environments exhibit challenging conditions including variable illumination, wave reflections, and occlusions from spray/mist. This dataset is leveraged for the near-surface Earth observation of coastal conditions.

Training was done using the NVIDIA TESLA P100, and inference was run using the NVIDIA 2080 and Intel 4-core XEON CPU. A batch size of 8 was used with the Adam optimizer [24] for 300 epochs.

On investigating the role of the Residual Block Configuration within LEMMA’s three-branch design, the authors

Model	mIoU	#Params (M)
R-LinkNet [7]	90.06	-
R-SGDBNet [15]	90.22	-
R-MANet [16]	91.92	26
R-FPN [28]	91.95	28
R-DeepLabv3 [9]	92	39.6
R-UNet [18]	92.04	24.5
R-Segformer [51]	92.04	62.6
R-DeepLabv3+ [10]	92.23	40
R-UPerNet [50]	92.33	30
R-GSSNet [8]	92.66	-
E-UPerNet	90.98	-
E-LinkNet	91.08	7.7
E-SegFormer [51]	91.2	3.7
E-DeepLabv3+	91.33	5
E-DeepLabv3	91.65	5
E-FPN	91.71	7.7
E-UNet	91.87	7.7
E-MANet	91.26	7.7
E-GSSNet	92.14	-
LEMMA (Ours)	93.42	1.01

Table 2. Comparison of various models with LEMMA on mIoU across the Oil Spill Drone [42] dataset. Parameters are provided for both ResNet-50 (R) and Efficient-Net-b1 backbones (E) where applicable. Results for LEMMA are of residual block (Section 3) configurations 6,7,4. Missing data is due to unreported values in original works or lack of publicly available weights. Parameters for some of the models have been calculated by the parameter count of backbones and other modules of the architecture as made available by authors.

found 7, 7, and 1 residual blocks (in the LFB, MFB, and HFB, respectively) for the MaStr1325 dataset and 6, 7, and 4 blocks for the Oil Spill Dataset to offer an optimal balance between model complexity and segmentation accuracy. While adding more blocks marginally improves performance, it would also increase computational cost.

Table 1 and Table 2 provide a consolidated view of the performance metrics: mean Intersection-over-Union (mIoU), trainable parameters, GFLOPs, and inference time. Comparisons are drawn against widely adopted CNN and Transformer-based approaches and specialised marine segmentation networks. Fig. 3 and Fig. 4 offer qualitative results illustrating how LEMMA parses fine-scale structures across marine environments.

As seen in Table 1, despite requiring up to 71x fewer trainable parameters than competing methods, LEMMA achieves comparable or better segmentation quality on the

Mastr1325 dataset. In particular, mIoU often falls within a 1–2% margin of heavier backbones with upto 88.5% decrease in GFLOPs at 84.65% lesser inference time. This is remarkable given that many of these larger models leverage massive pre-trained backbones (e.g., on ImageNet) or rely on Transformers for multi-level feature fusion. In contrast, LEMMA trains efficiently from scratch, leveraging the early-edge awareness inherent in Laplacian pyramids to bypass dense feature map computation, leading to around 1 Million parameters, with upto 17.83 GFLOPs and a fast inference time of 7.3 ms at its heaviest.

WODIS [11], a neural network specifically designed for water obstacle detection on automatic surface vehicles in the marine environment, achieves a mIoU of 91.3% at 43.2 FPS on the MaStr1325 test set. However, this result is 7.67% lower in mIoU and 68.47% slower than the proposed method. Similarly, WaSR-T, a transformer-based network achieves 99.80% mIoU, but requires about 71x more trainable parameters and 86.67% more GFLOPs. Even lightweight methods for segmentation such as LETNet [53] and UISSNet [21] lack by 14-15% mIoU and incur a heavier cost of upto 44x.

Moving to Table 2, LEMMA demonstrates better performance than various SOTA methods with various backbone selections. Here, the superior performance of LEMMA, at a much lesser cost of upto 62x is noted. The reduced mIoU in Table 2 as compared to Mastr1325 is primarily due to environmental factors such as reflections, waves, glare, and weather, which create inconsistencies in pixel intensities and impact the pyramidal representations, also visible in Fig. 4. Legacy SOTA methods like DeepLabv3, DeepLabv3+, UNet and recent methods like the GSSNet and MANet are outperformed by LEMMA, which showcases an mIoU of 93.42% at only 1.01M trainable parameters.

The most striking advantage of LEMMA is its ability to seamlessly adapt to contrasting viewpoints (surface level for MaStr1325 vs. aerial for Oil Spill Drone). Many state-of-the-art Earth Observation segmentation methods focus heavily on a single domain or incorporate specialised attention mechanisms and multimodal inputs to improve results, often at the cost of significantly higher complexity. LEMMA, on the other hand, uses a consistent, pyramid-based feature extraction that excels at highlighting thin boundaries in both small USV obstacle segmentation (e.g., floating debris or buoys) and large-scale UAV oil spill delineation (e.g., faint hydrocarbon sheens on water), showing cross-platform robustness. The Laplacian pyramids implicitly suppress low-frequency illumination drift caused by factors such as sun glint, water reflectance, and haze.

Qualitative Analysis in Fig. 3 and Fig. 4 showcase segmentation masks produced by LEMMA in both USV-based and UAV-based scenarios. The left column contains in-

put images, followed by results of UNet [41], DeepLabv3 [9], FPN [28], WaSR-T [58] and the proposed model, and the right column is the ground-truth mask. As seen across both datasets, the proposed methodology produces superior results for different frames of reference captured by aerial drones and surface-level sea vehicles. Additionally, LEMMA excels at segmenting not just oil spills but also obstacles like buoys, shorelines and piers at a much lesser computational load.

However, as noted in the last row of Fig. 4, a failure case is demonstrated. The authors believe this is primarily due to the content of the images. A large portion of the RGB image in Row 3 is covered by the reflection of the ship. Hence, in its Laplacian pyramid, due to the reflection, the intensity values of the water and the reflection are very similar, causing edges to be blurred and to be not as prominent. This leads our edge-guided model to inaccurately segment areas. These limitations of LEMMA as visible in the mask when compared to those generated by pretraining-dependent, computationally heavy models.

5. Ablation Study

In this section, we aim to evaluate the effects and analyze the trends observed through experiments involving varying residual blocks and loss functions for both datasets. This exhaustive experimentation justifies our choice of hyperparameters and enables our model to achieve state-of-the-art results with significantly fewer trainable parameters.

Table 3 illustrates the impact of the number of residual blocks in each branch of the model. As mentioned in Section 3, the number of residual blocks in the lower, middle, and higher branches are denoted as NRB_L , NRB_M and NRB_H , respectively. For optimal performance, the ideal configuration of residual blocks selected for the MaStr1325 dataset is 7, 7, 1, whereas for the Oil Spill dataset, it is 6, 7, 4. For both datasets, the optimal residual block configuration is determined by keeping the number of blocks in two branches fixed while varying the other. The highest mIoU of 98.96% is obtained with the 7, 7, 1 configuration for the MaStr1325 dataset and 93.42% is obtained with 6, 7, 4 on the Oil Spill dataset. Increasing the number of residual blocks in the top branch slightly reduces performance while increasing the number of parameters and GFLOPs, as it requires additional computations on larger feature maps. Similarly, the 6, 7, and 4 configurations are chosen for the Oil Spill dataset, where increasing NRB_L beyond 6 blocks slightly degrades performance, and increasing NRB_H up to 4 results in diminishing returns. This demonstrates that carefully chosen configurations can achieve high mIoU without unnecessarily increasing complexity.

Table 4 evaluates the effect of various loss functions. For these experiments, Focal Loss [29], Dice Loss [36], and

Dataset	NRB_L	NRB_M	NRB_H	mIOU
Mastr1325	3	3	3	0.9864
	3	4	3	0.9871
	3	5	3	0.9876
	3	6	3	0.9876
	3	7	3	0.9886
	4	7	3	0.9882
	5	7	3	0.9890
	6	7	3	0.9894
	7	7	3	0.9895
	7	7	1	0.9896
	7	7	2	0.9893
	7	7	3	0.9892
	7	7	4	0.9894
Oil Spill	3	3	3	0.9063
	3	4	3	0.9148
	3	5	3	0.9055
	3	6	3	0.9132
	3	7	3	0.9151
	4	7	3	0.9188
	5	7	3	0.9149
	6	7	3	0.9294
	7	7	3	0.9191
	6	7	1	0.9299
	6	7	2	0.9217
	6	7	4	0.9342

Table 3. Impact of Number of residual blocks in each branch of the model and the corresponding results on both datasets. The best results for every branch are bolded.

Dataset	Loss Function	mIOU
MaStr1325	Focal	0.9897
	Dice	0.9872
	CE+Dice	0.9886
Oil Spill	Focal	0.9342
	Dice	0.9262
	CE+Dice	0.9294

Table 4. Impact of various loss functions with best configurations of residual blocks on both datasets. The best loss function for each dataset is bolded.

Cross Entropy + Dice Loss are considered. The best performance on the Mastr1325 dataset is observed with Focal Loss, achieving an mIoU of 98.97%, while the Oil Spill dataset achieves its highest performance of 93.42% using Focal Loss as well.

6. Conclusion and Future Work

In this work, we introduced LEMMA, a novel lightweight semantic segmentation network for marine segmentation which bypasses heavy computational loads by employing

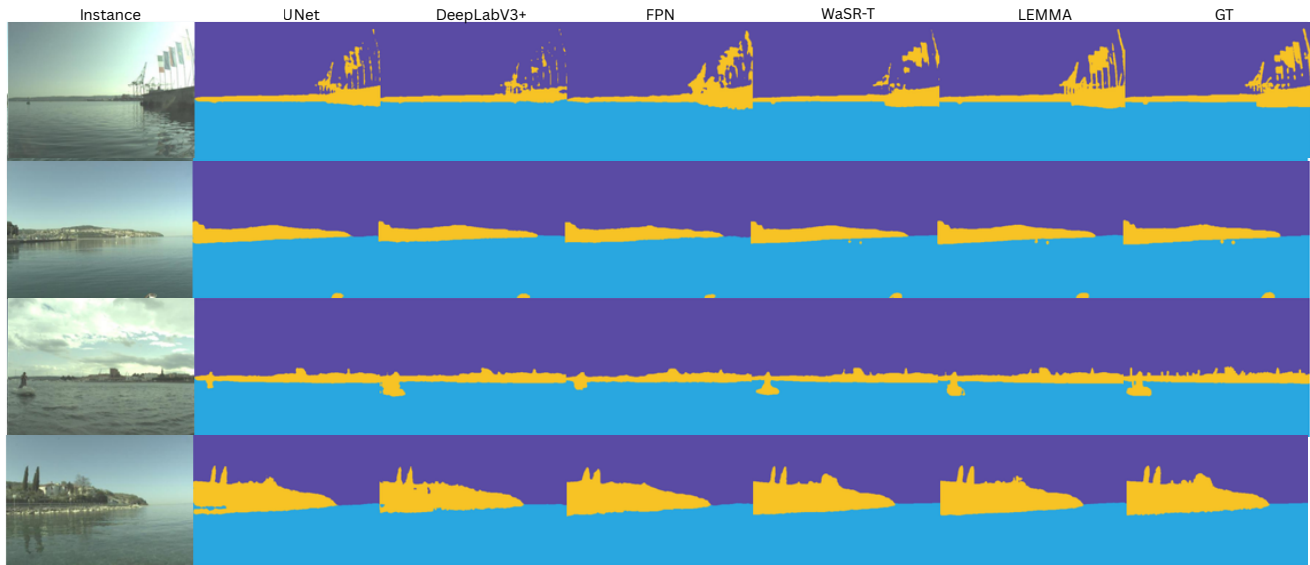


Figure 3. Visualization of the qualitative outputs on the Mastr1325 Dataset.

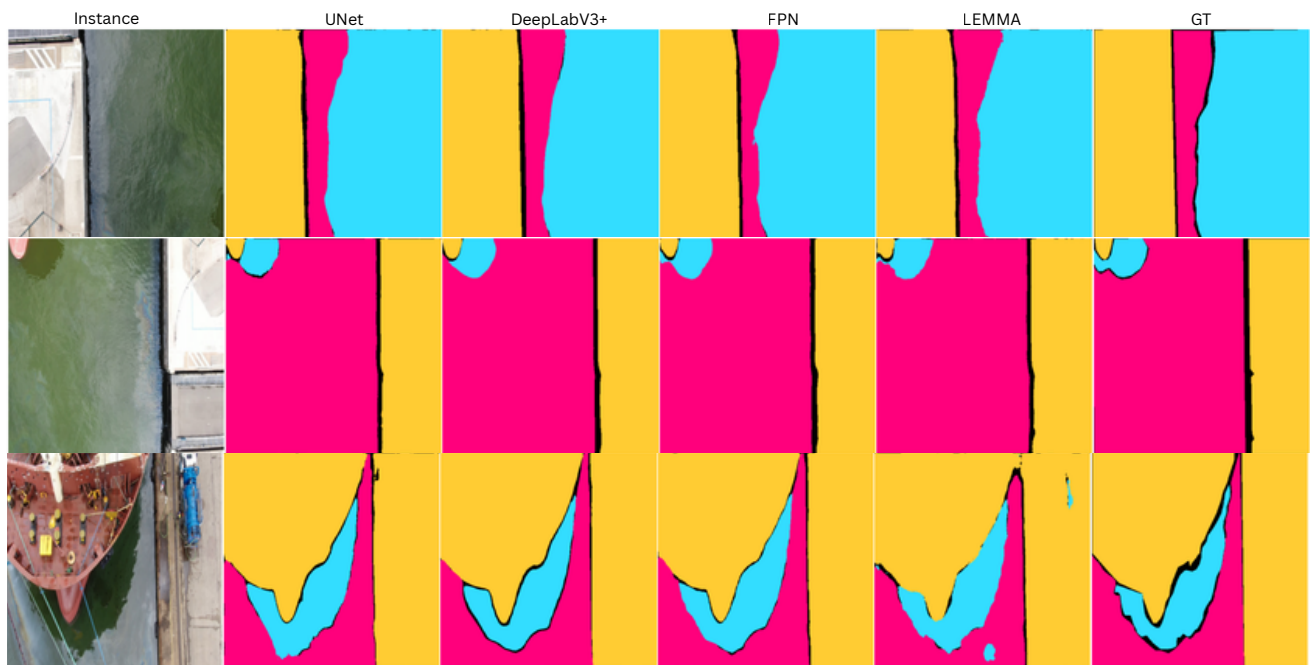


Figure 4. Visualization of the qualitative outputs on the Oil Spill Drone Dataset.

Laplacian Pyramids for feature extraction along with multi-level processing with residual blocks. LEMMA demonstrates comparable performance with up to 71x fewer parameters, making it deployable in real-world scenarios which need quick segmentation of obstacles and oil spills.

While LEMMA demonstrates promising results with significantly reduced parameter counts, several limitations remain. First, while the current datasets are standard in the

marine domain, they are limited in size and diversity. Limitations of this method include failure cases when reflections, waves, glare and environmental conditions are encountered. These affect the Laplacian pyramid and hence the subsequent construction of an accurate mask. Finally, LEMMA's current architecture uses fixed pyramid levels and static residual block configurations.

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